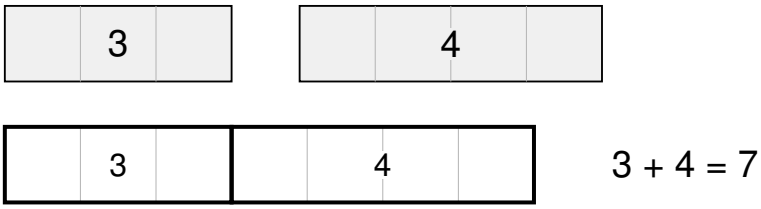
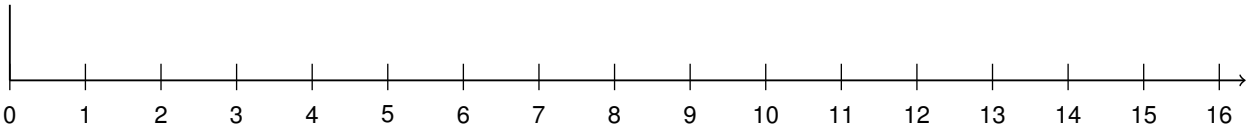
Put whole rods end to end. Leave no gaps or overlaps. You may use a length more than once. Changing the order of the same rods does not make a new way.




**Problem 1:** Use 3-rods and 4-rods. Find all the lengths from 1 through 16 that can be built, and mark the ones that cannot.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |



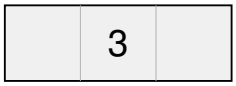
**Problem 2:** Find every way to make 12, 16 and 24 with 3-rods and 4-rods.

12

16

24

**Problem 3:** Use 3-rods and 5-rods. Find all the impossible lengths from 1 through 20. Is there a last impossible length?



| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 8 | 9 | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | |

**Problem 4:** Can 3-rods and 4-rods build every length after 16? Can 3-rods and 5-rods build every length after 20? Explain how you know.

**Problem 5:** Use 4-rods and 7-rods. Find the last impossible length, and explain why there are no later gaps.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 |
| 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 28 | | | | | | | |

**Problem 6:** Someone says every pair of rod lengths eventually has no more gaps. Is the claim true? Use rods 2, 3, 4, 5 and 7 to decide.